From Formal Authority to Practical Human Control

A traceable method for reconstructing human control in automated decisions

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Abstract

An assigned reviewer or approval step does not establish that a person could influence an automated decision. This study presents a documentary procedure for examining six conditions: information access, comprehension capacity, intervention authority, intervention feasibility, exercised judgment, and execution propagation. It applies the procedure to three purposively selected historical public-record packets. Each finding links a source, an interpretation, and a categorical state; missing evidence remains distinct from demonstrated failure. The preserved baseline reports two Fail results and one Unresolved result. A limited correction gives one Fail and two Unresolved results. A further, post hoc ZG710 reassessment gives three Unresolved results across 64 retained joint assignments, conditional on its approved judgments and the other limited-correction cells. Any justified necessary-condition failure still produces Fail. These revisions narrow the documentary claims; they do not establish that historical control was absent or improved. An open-index literature search places the method among existing control, reconstruction, and assurance approaches, with substantial coverage limits. The contribution is an inspectable implementation and evidence request, not an independently validated instrument. The three single-assessor cases establish no prevalence, reliability, causal effect, or transfer to contemporary AI systems. Institutions invoking human control should preserve records that can test whether assigned authority had practical force.

Keywords: human oversight; meaningful human control; AI governance; incident reconstruction; assurance

1 The question behind a human control claim

A person may be authorized to reject an automated recommendation yet lack the information, understanding, time, or working controls needed to do so. Conversely, agreement with a recommendation may follow a sound review. Neither a signature nor disagreement with the machine is, by itself, a sufficient test of practical control.

The distinction matters when an institution uses human oversight to justify deployment, reassure affected people, or allocate responsibility after harm. If a role assignment is mistaken for an effective safeguard, responsibility can be attached to someone whose authority had little practical force. Human-factors research explains how automation can leave people responsible for difficult exceptions while weakening the experience and situation awareness needed to handle them (Bainbridge, 1983; Endsley, 2017). Oversight policy can repeat the mistake by requiring human review without establishing that people can perform it (Green, 2022).

This paper asks: *How can a frozen, evidence-traceable procedure represent formal authority, practical human control, and unresolved evidence in one bounded public incident record?* Here, "bounded" means that the decision sequence and its start and end points are stated before the assessment. "Evidence-traceable" means that a reader can follow each conclusion back to the in-

spected source, the assessor's reasoning, and the rule used to combine findings.

The Trust Autonomy Evidence (TAE) procedure connects case selection, a versioned evidence packet, six condition-level judgments, and a case result. It does not determine personal blame, legal liability, or moral character. It also does not offer a new philosophical theory of meaningful human control. Related methods already provide retrospective public assessment, insufficient-evidence categories, practical-override traces, and executable assurance rules. The contribution is their particular integration into a revisable documentary assessment. Its uniqueness remains unverified.

Three historical packets demonstrate the procedure: the 1983 Oko early-warning episode and two 2003 Patriot engagements, ZG710 and F/A-18C. They were selected to supply different evidence conditions, not to represent a population or independently test classification accuracy. The central result is a correction of what these packets warrant. The historical baseline and subsequent reassessments remain separately visible throughout the paper.

How to read this paper. For the procedure, start with Section 4; for the case findings and disputed judgments, read Section 5. Figure 1 previews the three analyses. Section 8 separates what the study demonstrates from what remains untested.

2 What existing methods contribute

2.1 Defining control before looking for evidence

Meaningful-human-control accounts ask whether system behavior responds to relevant human reasons and can be traced to people who understand their responsibilities and the system's effects (Santoni de Sio & van den Hoven, 2018). This is a stronger requirement than human presence. Engineering accounts connect ability, authority, responsibility, and links between human decisions and AI actions (Siebert et al., 2023); purpose-sensitive accounts first ask what the control is intended to achieve (Davidovic, 2023). These approaches help identify whose control is claimed, over which decision, and through which options.

Control also operates beyond the immediate event. Design, deployment, institutional arrangements, and governance can enable or constrain intervention (Calvert, 2025; Verdiesen et al., 2021). Supervisory and teaming approaches describe different human-system relationships, while evaluative agency includes verification, steering, contestation, and substitution (Tsamados et al., 2025; Zhu et al., 2026). An event-level result therefore cannot settle whether the institution's whole control system is adequate.

2.2 Explaining why a formal role may not work

Human-factors research supplies mechanisms, rather than a shortcut from an outcome to a diagnosis. Automation can reduce practice; situation awareness can deteriorate; trust and reliance depend on the person, task, automation, and environment (Bainbridge, 1983; Endsley, 2017; Lee & See, 2004). Effective oversight also depends on epistemic access, causal power, self-control, fitting intentions, competence, and institutional incentives (Sterz et al., 2024; Laux, 2024). Corrective oversight requires actual preparedness and capacity, not participation alone (Baum & Laux, 2026).

A recorded approval or rejection does not identify its evaluative basis. Langer et al. (2025) conceptually distinguish sensitivity to errors from the tendency to classify outputs as erroneous. Across three Dutch experiments, Alon-Barkat and Busuioc (2023) did not find greater adherence to algorithmic than equivalent human-expert advice; selective adherence depended on context and was not reproduced in their civil-servant study. The conceptual distinction is not a new experiment. Together, these sources caution against treating agreement as evidence of one general automation bias.

Design proposals address the ability to question and act: counterargument prompts in medical support, ongoing human learning, and dynamic task-allocation patterns (Haselager et al., 2024; Te'eni et al., 2025; van der Waa et al., 2020). Legal analysis of AI agents likewise identifies speed,

Same packets, different warranted conclusions

	Released baseline v0.17.0	Limited correction	Additional ZG710 reassessment*
Oko	**Unresolved**	**Unresolved**	**Unresolved**
ZG710	**Fail**	**Fail**	**Unresolved**
F/A-18C	**Fail**	**Unresolved**	**Unresolved**
	0 / 2 / 1 Pass / Fail / Unresolved	**0 / 1 / 2** Pass / Fail / Unresolved	**0 / 0 / 3** Pass / Fail / Unresolved

* Conditional on retained judgments; 64 joint assignments. Not a control score.

Figure 1. Documentary conclusions across three analyses of the same selected packets. Each column is a set of assessment judgments, not a later historical event, a separate experiment, or a measured improvement in control. The additional ZG710 comparison is conditional on its retained judgments and the other limited-correction cells: all 64 retained joint assignments give three Unresolved results. Any justified necessary-condition U still produces Fail. Table 4 preserves the complete condition-level states.

opacity, and monitoring and recordkeeping constraints on oversight (Kolt, 2025). These are reasons to inspect the conditions of a decision, not findings that the conditions were present in a particular case.

Recent studies illustrate the distinction. Developer interviews describe advance controls, coplanning, real-time monitoring, and later review in software-agent work (Dhanorkar et al., 2026). A secondary analysis of 99 Tesla-user interviews concerns perceived control, not observed incident-control outcomes (Suryana et al., 2024). In a 28-participant sandbox study, task completion was 83.3% under direct manipulation and 61.9% under text guidance, with delays and difficult error correction complicating intervention (Li et al., 2025). Its sample and tasks limit transfer. None of these sources makes an available control or recorded action sufficient proof of effective intervention.

2.3 Reconstructing an event from incomplete records

Reconstruction asks what people could know and why their actions made sense as the event unfolded, rather than assigning hindsight-based blame (Dekker, 2002). McDermid identifies sufficient time, independent information, and requisite skills as prerequisites for intervention (McDermid, 2019). Method and source coverage also shape the result. In a comparison of HFACS, AcciMap, and STAMP applied to one official accident report, the methods produced different factor sets; missing report detail could hide relevant factors (Zhang et al., 2018). That comparison did not validate any method against complete event truth.

The AI Incident Database supports cross-incident learning, but its public reports vary in coverage and detail (McGregor, 2021). Classification and editorial work already recognize uncertain causes and structural ambiguity (Pittaras & McGregor, 2023; Paeth et al., 2025). Public inquiries can support sociotechnical analysis, while agent-incident research identifies useful logs, system documentation, access, and tool records (Macrae, 2022; Ezell et al., 2025). Reporting institutions determine which evidence becomes available and what follows from it (Wei & Heim, 2026). These precedents support an explicit indeterminate category and visible source dependencies.

2.4 Making the inference inspectable

Argument-based assurance connects a claim to evidence under stated assumptions. Ethical assurance, machine-learning safety assurance, and algorithmic assurance audits apply that discipline to different claims and settings (Burr & Leslie, 2023; Paterson et al., 2025; Lam et al., 2024). This paper borrows the discipline without issuing a safety case for Oko or Patriot.

Evidence-linked research and argument graphs also precede this work. ScientistOne links claims and evidence and audits scores, specifications, references, and method-code alignment; some judgments are model-based and human-reviewed, so the whole process is not deterministic (Meng et al., 2026). Moghaddam (2026) proposes admissibility, completeness, provenance, and repair checks. Prospective governance frameworks connect oversight and record duties (Alhalangy, 2026; Lee et al., 2026). Forensic proposals address changing system state and prompt-level evidence preservation (Leung et al., 2026; Ledjaki et al., 2026); only their abstracts were inspected, so the citations support their stated purposes, not effectiveness claims.

The closest procedural overlap is substantial. CEDAR-42001 combines retrospective public evidence, an insufficient-evidence category, and versioned executable rules (Surve et al., 2026). Niazi et al. (2026) describe versioned assurance cases, practical-override traces, and retrospective process tracing. LaPosta (2026) proposes a runnable human decision-chain protocol in a one-page extended abstract; Bahidika (2026) maps incident annotations into oversight tiers. Retrospection, traceability, and executable assessment are not new by themselves. The present study demonstrates a specific path from selection to packet, provenance, condition state, and revisable conclusion. It does not establish independent agreement or institutional benefit.

2.5 A compact comparison and a source dependency

Table 1 distinguishes the questions emphasized by four approaches. It is a reading aid, not a ranking or a claim that their full scopes are interchangeable. The McCarthy comparison is a contextual addition from the separate 8 November reading review. It does not revise the frozen literature-search population, add evidence to a historical packet, or change a case label.

McCarthy's (2013) MIT thesis applies CAST to the same Tornado event, examining training, communications, and relationships within the wider system. Its event sequence draws on the MoD inquiry summary and also uses the Defense Science Board report. TAE draws on overlapping material. Two analyses of the same summary do not make two independent factual witnesses: the thesis is related analysis, not independent corroboration of every event fact or of TAE's conclusions. The focused reading reviewed selected chapters and bibliography, not a new systematic corpus or complete artifact audit.

3 Study design and case selection

3.1 Scope and sequence

This is a retrospective, purposefully selected, single-assessor methods study. The unit is one bounded machine-mediated decision or action sequence. The functional inclusion rule covered systems generating an inference, classification, recommendation, communication, decision, or action with consequences for people or their settings. Oko is a rule-based early-warning system; Patriot combines automated detection, tracking, classification, and engagement support. Neither is equivalent to a present learned model or agent. Their public records make the sequence examinable, but any transfer depends on shared mechanisms and a new test with contemporary traces.

The original protocol and empty selection register reached the repository's main branch at commit `180ddda1d70f0ee36faaf8875e839bbc99cbbec2`. Screening began afterward. The evidence cutoff was **6 August 2026 at 23:59:59 UTC**. The protocol fixed two collections, eighteen search terms, six eligibility conditions, exclusions, chronological order, three strata, the first-eligible rule, and missing-data treatment. This freeze constrains the original selection, not the later post hoc reassessment. The 5 September policy review and 10 October source and manuscript review are separate checkpoints, not additions to the

Table 1. Related approaches and the question emphasized in this comparison

Approach	Question and useful contribution	Limit of the comparison
McCarthy's CAST analysis	How did the wider control structure and constraints contribute to the ZG710 event? Systems-safety reconstruction extends beyond one person's actions.	Later analysis of records, not independent observation of the event or proof of individual cognition.
CEDAR-42001	How should audit obligations, evidence judgments, and improvement priorities be organized? Insufficient evidence can remain Not assessed.	Repeatable processing of supplied coding is not independent agreement about that coding.
Niazi and colleagues	How can rights, risks, rules, and control evidence remain connected? Versioned assurance cases and override traces link claims to records.	Proposed mechanisms do not establish deployment or validation; underlying coding artifacts were not independently checked here.
Present TAE procedure	What does the bounded record support about six conditions of practical human control? Source, interpretation, state, and correction remain inspectable.	Three selected, single-assessor packets do not establish reliability, superior accuracy, or institutional benefit.

historical packets.

3.2 Search boundary and stopping rule

The AI Incident Database snapshot contained 1,607 incidents and 7,452 linked reports; 828 incidents matched at least one frozen term. The OECD AI Incidents and Hazards Monitor exposed the first 100 rows of the frozen query, despite reporting about 3,635 matches. The combined output therefore contained **928 candidate records**, not an unrestricted census of the two collections. Snapshot and export hashes were recorded. Date probes found no OECD result before 2020; all selected cases predated 2020 and came from the AI Incident Database.

Eligibility required an identifiable machine output or action, a bounded sequence, two reports from different authors or organizations, a primary or official record, lawful citation and analysis, and cutoff compliance. The incomplete-evidence stratum permitted two independent secondary reports after a documented unsuccessful official-record search. Exclusions covered benchmarks, unbounded performance reports, single-source events, dependence on private evidence, rights conflicts, duplicates, and post-cutoff evidence.

Candidates were deduplicated, ordered by event date and collection identifier, and screened until the first eligible case filled each stratum: intervention before irreversible action, authority without practical force, and incomplete or conflicting evidence. Five candidates were screened. Two were excluded, one for insufficient independent reporting of a bounded action and one for no bounded action sequence. AIID-27, AIID-444, and AIID-445 filled the strata, respectively. The remaining **923 candidates were unscreened**, not excluded (Figure 3).

3.3 What a packet preserves

Each packet defines the action boundary and chronology; records source titles, authors, issuers, dates, URLs, classes, and preservation status; identifies unavailable evidence; labels material statements; and states publication-rights limits. Local files have hashes; remote-only sources have metadata and retrieval information. A hash identifies stored bytes, not source truth, event authenticity, or later remote content.

The packet also records dependencies. A later inquiry can describe contemporaneous logs that this study did not inspect. Reports that draw on the same inquiry or participant are not independent confirmations merely because they have

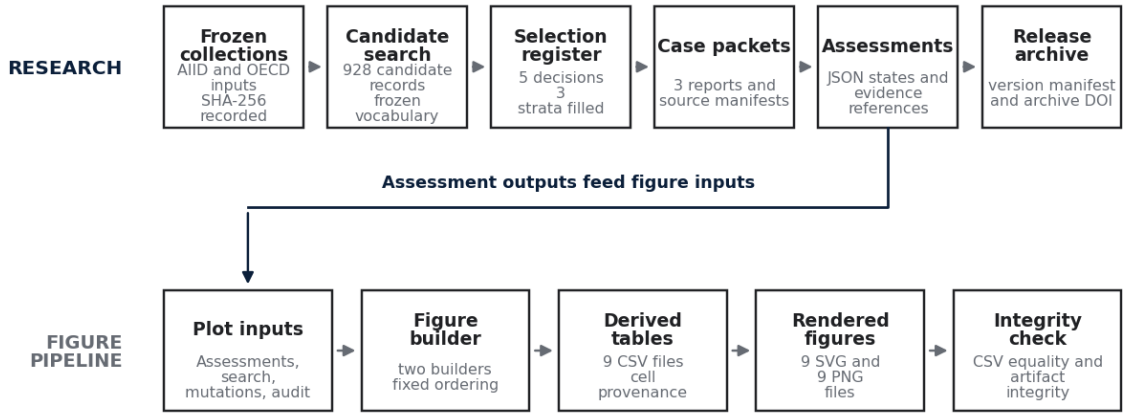


Figure 2. Study record and reproducibility path, retained from the released manuscript. Frozen collections lead to selection, packets, assessments, and an archive; recorded states supply the figure inputs. Checks can reproduce these transformations, but source interpretation remains an assessor judgment. The later reassessment supplement is a separate versioned derivation, not a replacement for this historical pipeline.

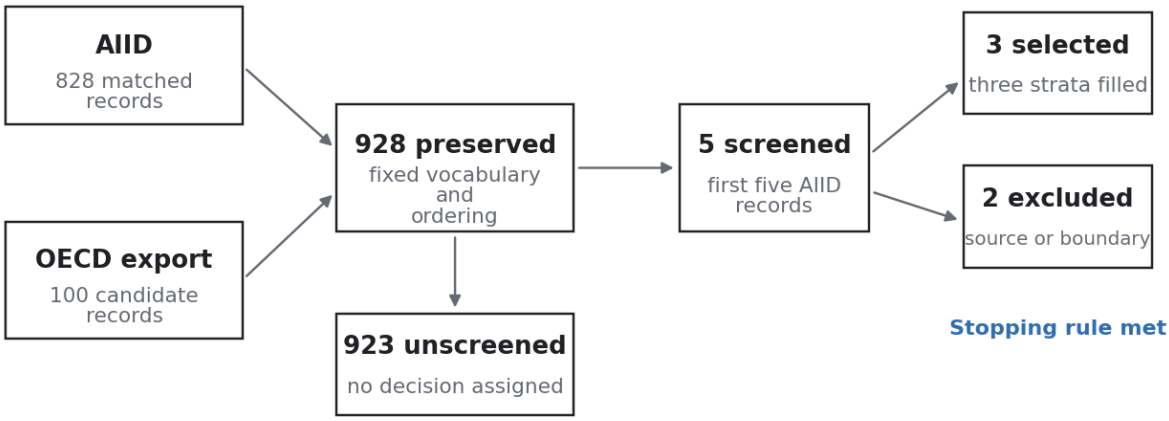


Figure 3. Frozen selection and stopping. The preserved five decisions filled three prespecified strata. Because those strata anticipated the desired contrasts, this selection demonstrates execution under chosen evidence conditions, not independent discrimination or prevalence.

different publishers. This matters particularly for Oko's retrospective accounts and for the Patriot inquiries available only through summaries or reporting.

4 How the assessment works

4.1 Six necessary conditions

The assessment follows a practical question: what must hold for assigned authority to affect execution? It starts with the information available to

the person and ends with the system or institution carrying out the person's judgment. Table 2 gives the six questions and the kinds of records that can address them. A record type is not automatic proof; it must support the specified proposition at the declared boundary.

Two distinctions are essential. First, *execution propagation* asks whether a judgment or intervention was implemented, not whether its outcome was beneficial. A harmful action can be faithfully executed. The released JSON field named "effect" refers to this propagation question. Second, *exercise* asks for active judgment or intervention. Departure, challenge, and escalation are stronger evidence, not an unambiguous universal requirement. Automatic approval and retrospective rationale remain failure indicators. The reassessment neither demands independent challenge in every case nor awards partial support for any recorded action.

4.2 States and provenance

Each required condition receives one of four states (Table 3). Outside scope is available only for a justified nonrequired proposition; it cannot remove any of the six event conditions. These categories have no numerical distance, and the procedure defines no summed control score.

A missing public record does not itself show that a condition was absent. An absence inference needs adequate evidence about what should have been recorded and the coverage actually inspected. The direct-contemporaneous requirement applies to S. The reassessment does **not** create a matching contemporaneity gate for U or automatically discard later adverse findings. The original rules are less explicit about when retrospective evidence establishes a component for P; unresolved choices remain visible rather than being silently resolved.

Material statements are separately labeled *direct record*, *source claim*, *assessor inference*, or *unresolved*. For example, a report of deficient training is a source finding. Concluding that it establishes this crew's failure of a necessary capacity is an inference. Deriving Fail after the assessor assigns U is a calculation. Software can verify the last step without validating either of the first two.

A hypothetical example. A log says that a reviewer clicked "approve." That is evidence of an action, not by itself evidence that the reviewer understood the output, could stop it in time, or that the approval reached execution. Those are separate questions. The example illustrates the distinctions; it is not another study case.

4.3 The case rule and unresolved alternatives

For case c, let $x_{c1}, \ldots, x_{c6} \in \{S, P, U, I\}$ be the six required states. Then

$$\text{EventControl}(c) = \begin{cases} \text{Fail}, & \text{if any } x_{cj} = U, \\ \text{Pass}, & \text{if every } x_{cj} = S, \\ \text{Unresolved}, & \text{otherwise.} \end{cases}$$

The order is deliberate: **an established necessary-condition failure has priority**. Uncertainty elsewhere cannot cancel U. Pass requires six S findings; all remaining combinations are Unresolved. Correction and repair are additional requirements for a wider institutional-accountability claim, and reform requires longitudinal evidence. Those post-event judgments are not part of the six-condition aggregate.

The notation $\{P, I\}$ means that P and I remain alternative classifications requiring adjudication. It is not a fifth event state or a blended score. Enumeration asks whether choosing either retained state changes the case result. An invariant result does not decide which label the evidence warrants or establish that the retained alternatives are correct.

4.4 Checks and the scope of reassessment

Repository checks cover schemas, packet hashes, selection invariants, cross-case interactions, figure inputs, release manifests, and prescribed corruptions. The Chain-of-Evidence adaptation links claims to exact source locators and records support attestations and five evidence-fitness dimensions: directness, contemporaneity, independence, completeness, and publication authority. A dependency graph restricts which claims may support conclusions. These controls detect specified inconsistencies and reproduce declared calculations.

Table 2. The six event conditions and their documentary tests

Condition	Question	Relevant record
Information access	Did the person receive the information needed for the claimed review?	Timestamped delivery, interface, or access evidence.
Comprehension capacity	Could the person interpret the information and relevant limits under the operating conditions?	Role-appropriate materials and evidence of usable capacity.
Intervention authority	Was the person permitted to change, reject, stop, or escalate the decision?	Delegation and actual control rights.
Intervention feasibility	Could that authority be used in time under the actual conditions?	Time, staffing, access, and a workable response path.
Exercised judgment	Did active judgment or a qualifying intervention occur?	Contemporaneous decision, review, or communication evidence.
Execution propagation	Did the judgment or intervention reach execution or alter obligations?	A linked action, stop, modification, or escalation event.

Table 3. Condition states and decision rules

State	Code	Rule
Supported	S	Direct, contemporaneous evidence satisfies the condition.
Partially supported	P	Some required elements are established and a material gap remains.
Unsupported	U	Available evidence contradicts the condition or shows its absence, including failure of a necessary component.
Indeterminate	I	The packet does not contain enough evidence to decide.
Outside scope	O	A written boundary justification excludes a nonrequired proposition.

They do not independently validate historical interpretation.

Three comparisons use the same aggregation rule: the released v0.17.0 baseline; a limited correction of two harm-based propagation rationales and two Patriot authority S findings; and an additional ZG710 reassessment of comprehension, feasibility, and exercise. The latter is conditional on the other limited-correction cells. These changes were post hoc, author-led, and AI-assisted. They added no historical source or independent human assessor. Retrospective S, shorter case boundaries, and a new action-plus-evaluation exercise construct were not adopted.

5 Historical findings and their reassessment

5.1 What changes across versions

Table 4 gives every six-condition row and its derived result. The older Oko v0.3.0 all-S assessment predates the baseline shown here: a frozen v0.6 adjudication had already changed it to six P. That correction remains separate from the Patriot changes.

5.2 Oko in 1983

The released Oko boundary begins when the warning system classified incoming missiles and ends when Stanislav Petrov reported the alert as false before it advanced to a retaliatory launch decision. This is the assessment's attributed retrospective account. It is not a finding that Petrov suppressed every alert or alone prevented retaliation.

The packet contains participant accounts published later and a research synthesis drawing on related participant and interview sources (Hoffman, 1999; Lewis et al., 2014; Shuster, 2017). It contains no located contemporaneous Soviet command log, official incident record, or investigation file. Later first-person testimony directly supports what the participant later reported, but does not

Table 4. Preserved baseline and separately identified reassessments

Analysis and case	Access	Comprehension	Authority	Feasibility	Exercise	Propagation	Result
Released v0.17.0 baseline							
Oko	P	P	P	P	P	P	Unresolved
ZG710	P	U	S	U	U	U	Fail
F/A-18C	P	I	S	I	I	U	Fail
Limited correction							
Oko	P	P	P	P	P	P	Unresolved
ZG710	P	U	$\{P, I\}$	U	U	$\{P, I\}$	Fail
F/A-18C	P	I	$\{P, I\}$	I	I	$\{P, I\}$	Unresolved
Additional ZG710 reassessment, conditional on the other limited cells							
Oko	P	P	P	P	P	P	Unresolved
ZG710	P	I	$\{P, I\}$	$\{P, I\}$	$\{P, I\}$	$\{P, I\}$	Unresolved
F/A-18C	P	I	$\{P, I\}$	I	I	$\{P, I\}$	Unresolved

Access = information access; comprehension = comprehension capacity; authority and feasibility concern intervention; exercise = exercised judgment; propagation = execution propagation. $\{P, I\}$ retains two alternatives, not a combined state. Totals in Pass/Fail/Unresolved order are 0/2/1, 0/1/2, and conditional 0/0/3. The supplement retains the original baseline and limited-correction graphics and unreassessed post-event rows.

satisfy the contemporaneous-record requirement for S about the 1983 event.

The frozen v0.6 adjudication therefore changed all six v0.3.0 S findings to P with no new historical source. Oko remains P/P/P/P/P/P in all three comparisons, producing Unresolved. The packet supports a bounded account of later reporting, not a sole-cause counterfactual about retaliation. Supplement Section S1 preserves the six missing-record categories and original correction display.

5.3 Patriot ZG710 in 2003

The ZG710 boundary runs from classification through launch and destruction. Here, the assessed "crew" is the U.S. Patriot battery crew, not the RAF aircrew killed in the engagement. The assessment concerns the Patriot crew's authority and actions, not blame or legal liability. An Unresolved classification does not overturn the inquiry's adverse findings or diminish the loss of life. Crew-level evidence can address that proposition, but information distributed across people cannot establish an individual's cognition. The main sources are the inspected official inquiry summary T2 and Hawley's later technical account T4. Their timing and threat-classification descriptions are not silently reconciled.

The baseline P/U/S/U/U/U row gives Fail. The limited correction changes authority S and propagation U to $\{P, I\}$, while retaining the three other U cells, so the result remains Fail. The additional reassessment changes comprehension to I and leaves feasibility and exercise as $\{P, I\}$. The reasons for those three changes must be examined separately from their common numerical consequence.

Comprehension. The inquiry reports preparation oriented toward generic threats and deficient preparation for the circumstances and false-alarm problem (T2, section 11, printed p. 3). Hawley reports that known classification fallibilities were not incorporated into training and practices, and discusses problematic user beliefs (T4, p. 4). The strongest argument for U is that these findings establish the crew's lack of usable understanding of relevant limitations. If that necessary condition failed, routine operational ability or uncertainty elsewhere could not cancel it.

The additional reassessment judges that event-level link insufficiently established. Deficient preparation is evidence of an institutional capacity-development weakness, but is not itself a demonstrated account of this crew's comprehension. Hawley's model is broad, not an operator-

by-operator reconstruction, and the inquiry does not resolve the link through demonstrated interpretation. The operational sequence also supplies no positive capacity component by itself. The disposition is I: insufficient evidence to decide, not a finding that the crew understood the system or that the adverse evidence was false.

Feasibility. The inquiry reports limited time, communications equipment in transit, a relay through another battery, an incomplete wider air picture, and identification constraints (T2, sections 11–12 and 14, pp. 3–4). Together these could show that an indispensable information or response function was unavailable in time. A U finding would require identifying that necessary function and showing that no timely adequate route existed. It need not prove that every imaginable action was impossible.

The reassessment finds the function-to-route inference insufficiently established in the inspected packet. A crew action, delay, or radio relay can identify a candidate resource, but a functioning engagement route does not prove feasible informed challenge. Timing remains unresolved: T2 reports about one minute; T4 names the Tornado in a less-than-ten-second discussion of tactical-ballistic-missile timelines, whereas T2 describes an anti-radiation-missile classification. The endpoints and category mismatch are unreconciled. No complete required-response comparison can be calculated. P is retained only if a specified resource or route establishes a genuine component of the original informed intervention, with a material gap. Otherwise I applies. Neither alternative establishes that effective challenge was feasible.

Exercise. T2 section 4 attributes the launch to the crew. Hawley's broader account of automation misuse and uncritical acquiescence supplies adverse evidence (T4, pp. 4–5). U would follow if this account were sufficiently attributed to the decision and established failure of the original active-judgment-or-intervention criterion. Relabeling a launch "intervention" cannot evade the automatic-approval failure indicator.

That event-specific inference remains insufficiently established in the reassessment. Lack of independent challenge alone does not settle the original criterion, which permits active judgment or intervention and treats departure as stronger evidence. Yet a crew-attributed launch does not automatically earn P. P requires a reason why the action supplies a qualifying intervention component and a statement of the remaining gap; I applies if neither active judgment nor qualifying intervention has a supported component. Missing evaluation alone cannot establish U while a qualifying intervention branch remains. A demonstrated failure of the applicable criterion still would.

These are contestable source-to-condition judgments, not new evidence of competence, opportunity, or meaningful intervention. The adverse interpretations identify what would reverse them. The additional row is P/I/$\{P, I\}$/$\{P, I\}$/$\{P, I\}$/$\{P, I\}$, conditional on the other limited-correction cells.

5.4 Patriot F/A-18C in 2003

The F/A-18C boundary includes detection, classification, engagement, and aircraft destruction; it is not narrowed to an order. An inquiry-based account connects identification, ordering, engagement, and loss (Ricks, 2004, F3). The full inquiry, system logs, operator displays, and classified technical report were unavailable. The contemporary initial report supports the reported incident and intended inquiry, not the later detailed command trace (Burnham, 2003). The Defense Science Board summary supplies context and acknowledges causal limits, not an independently inspected operator trace (Defense Science Board, 2005).

The baseline P/I/S/I/I/U row gives Fail. In the limited correction, authority becomes $\{P, I\}$ because the packet lacks an inspected original rights record meeting S. Propagation also becomes $\{P, I\}$ because aircraft loss alone does not establish non-propagation. The corrected P/I/$\{P, I\}$/I/I/$\{P, I\}$ row gives Unresolved across four assignments and is unchanged in the additional comparison.

The same linked-action threshold must govern both Patriot cases. T2 section 4 attributes the ZG710 launch to its crew within the full event account. F3's main paragraphs 1 and 3–7 must be read in context through paragraph 9, rather than reducing the evidence to one ordering sen-

Table 5. One worked distinction between source, judgment, and calculation

Step	ZG710 comprehension example
Source finding	T2 section 11 reports preparation deficiencies; T4 p. 4 reports missing incorporation of known classification fallibilities into training and practices.
Assessor question	Do those findings establish that the incident crew lacked a necessary capacity to understand the relevant limitations? They are adverse evidence, not merely missing records.
Retained judgment	Additional reassessment: I, because the event-level capacity link is judged insufficiently established. This is contestable and is not proof of comprehension.
Calculation	With no U elsewhere in the retained additional row, the case is Unresolved. If a justified U replaces this I, the case becomes Fail.

tence. A defensible P reading of F3 treats the complete account as evidence of a human-decision-to-execution connection. This remains an assessor inference, not an inspected command acknowledgment. If it requires an unavailable premise or fails the required specificity, I follows.

The directly inspected official ZG710 summary and the less directly available inquiry basis behind F3 differ in evidentiary access. That difference matters to a particular inference; it does not justify accepting narrative attribution in one case while demanding a complete original trace in the other. Neither a lone order nor a coherent narrative automatically proves propagation. The unresolved eligibility of later evidence for P remains visible for both cases.

5.5 What the 64 assignments establish

ZG710 has four P/I choices, giving $2^4 = 16$ assignments. F/A-18C has two, giving $2^2 = 4$. Their Cartesian product contains $16 \times 4 = 64$ joint assignments. Oko stays six P. Under every retained assignment, no case contains U and none consists entirely of S, so all three case results are Unresolved (Figure 5).

This is a calculation over supplied state sets, not a sensitivity test over every possible coding rule or historical interpretation. It does not resolve the six P/I choices. Across eighteen event-condition findings, the additional comparison contains eight to fourteen P and four to ten I, with paired counts summing to eighteen and no S or U. These are classification descriptions, not prevalence estimates.

The separately versioned checker reproduces the historical and limited inputs, enumerates the alternatives, and preserves the failure-priority behavior. It does not decide whether the sources warrant a U, P, or I. The original trust-evidence matrix and post-event findings remain unreassessed secondary results in the supplement. They define no validated aggregate trust score or case ranking and cannot silently substitute for a six-condition result.

5.6 Literature search and integrity results

The formal search used a frozen protocol with eight query families, inclusion and screening rules, deduplication, fifteen citation seeds, and a contribution rejection test. Semantic Scholar was the controlling reproducible index; Crossref checked DOI metadata and OpenAlex compared seed coverage. Source and proposition review used publisher, proceedings, preprint, and institutional pages. Comparator software was not executed.

The eight queries returned 184 records; fourteen resolved seed chains returned 2,482 references and citations. L12 returned 404 at both chain endpoints. Deduplication left **2,431 records**. The initial AI proposals and 77-record attention queue produced an 89-record author-review queue. All twelve proposed close records were retained; the attention queue yielded fifteen more close, 32 background, twenty topic exclusions, and ten single-component exclusions. The final full-pool classes were 27 close, 45 background, 1,259 topic exclusions, ten single-component exclusions, 1,087 inaccessible, and three outside the cutoff (Figure 6).

A later frozen 102-record forward-citation recovery stratum yielded 71 records with content, all

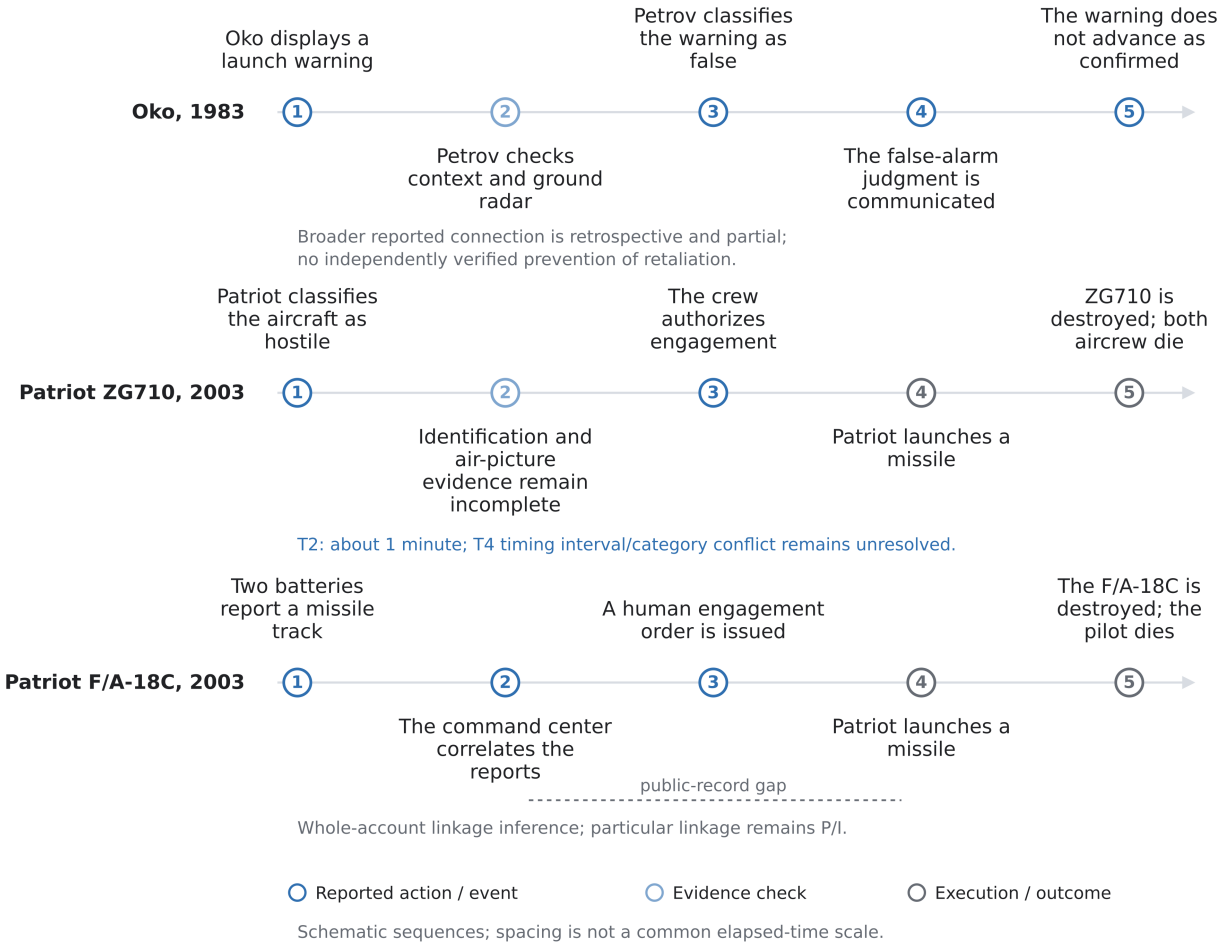


Figure 4. Bounded decision paths, retained from the November manuscript. Events are spaced for reading, not on a common time scale. Oko remains a retrospective, partially supported account. The Patriot paths do not equate harm with failed propagation; source attribution and uncertain links remain explicit. The figure supplies no causal-effect estimate or retaliation-prevention counterfactual.

Table 6. Conditional aggregation and the failure-priority check

Inputs held or changed	Assignments	Consequence
Limited-correction F/A-18C row	4	Unresolved in all four.
Additional ZG710 row	16	Unresolved in all sixteen.
Joint additional comparison	64	0 Pass, 0 Fail, 3 Unresolved.
Restore any justified U in ZG710 comprehension, feasibility, or exercise	Rule check	ZG710 becomes Fail, regardless of other P/I choices.

screened with authorized AI assistance. Thirteen entered close-source review; only five received bounded proposition permission, two remained background-only, and six were quarantined. Across the original 1,087-record recovery population, 107 retrieval outcomes and 76 recovered-content screening decisions were recorded; **980 records still lacked a retrieval outcome**. The remaining outcomes include 26 metadata-only, two unavailable, and three reconciled duplicates. RS-DQ-004 has a recorded screening decision but zero source-content per-

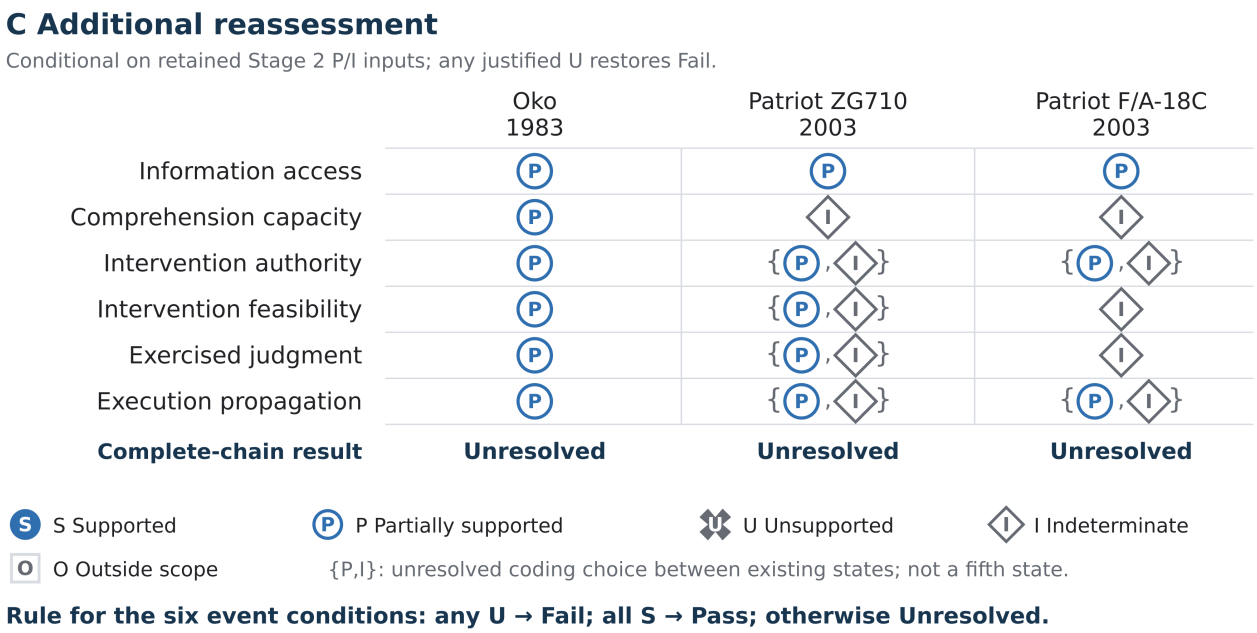


Figure 5. Additional ZG710 reassessment, retained from the November manuscript. The displayed conclusion is conditional on the approved additional judgments and the other limited-correction cells. Braces preserve unresolved P/I choices. Any justified necessary-condition U restores Fail.

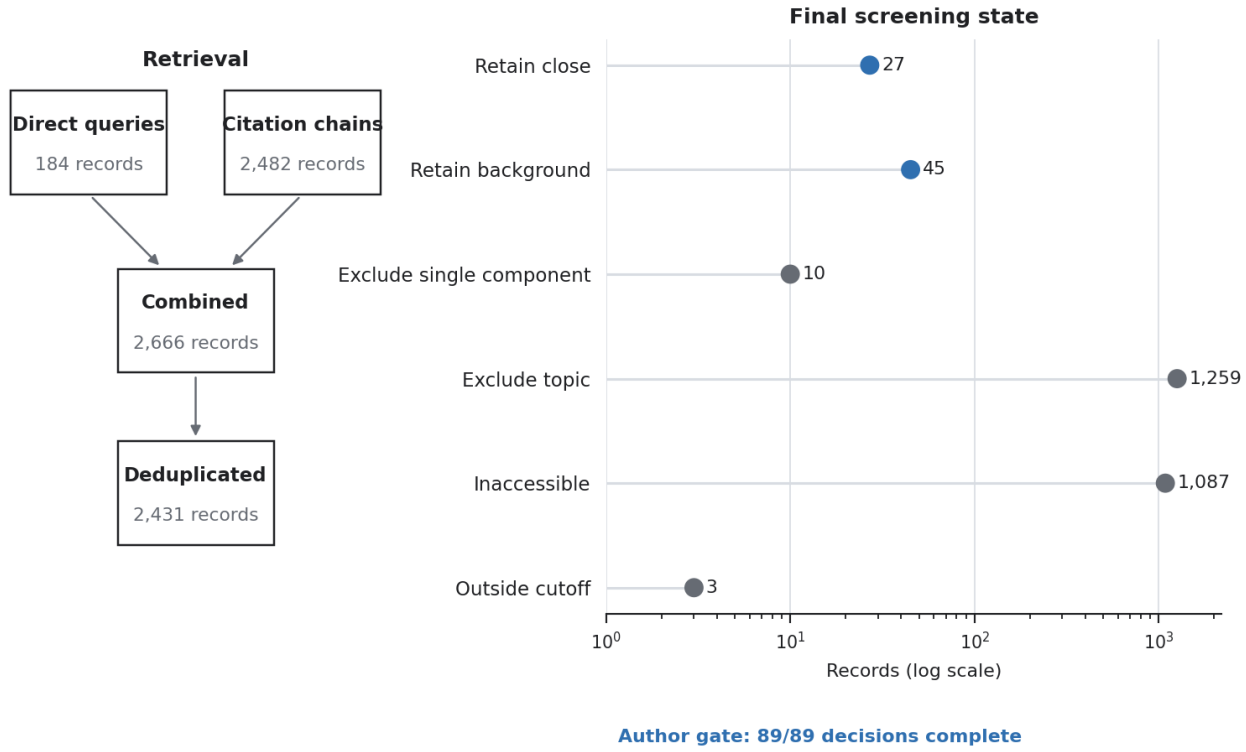


Figure 6. Preserved search retrieval and screening counts. The right panel uses a logarithmic count axis. Closure of the 89 author decisions does not close inaccessible-content or database-coverage limits. Later recovery outcomes remain separately recorded rather than silently replacing this checkpoint.

mission. These are distinct workflow counts, not estimates of close-source prevalence.

The audits also have separate scopes. The v0.16.0 base mapped forty claims, found 37 eligible within their stated limits, and detected all 39 prescribed corruptions. The v0.17.0 policy extension mapped five claims, all boundedly eligible, and detected nine prescribed corruptions. They are not a pooled validation rate. Earlier counts, exceptions, and displays are preserved in the supplement. A seeded-error check shows response to that error, not sensitivity to unknown mistakes. None of those audits independently evaluated the later source judgments.

6 What can be concluded from the records

Formal authority answers who was permitted to change the decision. In this procedure, the practical-control assessment additionally asks whether the person had usable information and capacity, a feasible opportunity, exercised judgment or intervention, and a connection to execution. The historical baseline illustrated this distinction by pairing S authority with Fail in both Patriot cases. The current reassessment no longer supports presenting that pairing as the current cross-case finding: authority itself remains P/I, and the additional comparison is conditionally Unresolved for all three cases.

An Unresolved result does not establish either that control existed or that it was absent. Public records are shaped by classification, litigation, disclosure, preservation, and time; evidence may exist elsewhere. An investigator should not infer that a person lacked capacity merely because the needed display or inquiry is unavailable. Conversely, uncertainty is not a reason to ignore adverse evidence that establishes a necessary failure.

The institutional concern is an unsubstantiated claim. Where an institutional assertion of human control is itself evidenced, the proposed documentary indicator is

$$\mathrm{EvidenceGap}(c) = \mathbf{1}\begin{bmatrix}\mathrm{HumanControlClaim}(c) = 1 \\ \wedge\, \mathrm{EventControl}(c) \neq \mathrm{Pass}\end{bmatrix}.$$

This conditional device identifies a claim not fully substantiated under the procedure. It is not a validated safety, honesty, or control measure. These historical packets do not automatically establish that an institution made the relevant assertion. Neither Fail nor Unresolved alone establishes personal fault, deliberate misrepresentation, legal liability, or an unsafe outcome.

The ethical proposal concerns the institution's evidence, not a presumption against an operator. If human-control language helps justify deployment or assign responsibility, records should permit that claim to be tested. This follows established concerns about contestability, institutional design, and the mismatch between responsibility and practical power (Almada, 2019; Laux, 2024; Douer & Meyer, 2020). The proposed evidence request translates those concerns into inspectable questions; it guarantees no safe or just outcome.

7 Prospective use and what would need validation

7.1 An evidence request rather than a compliance verdict

The relevant record set follows from the claim, actor, options, and boundary. A reviewer could request the information and system output actually shown; linked output, review, action, and commitment times; system version and known limits; independent information needed for challenge; delegation and tested control rights; staffing, workload, latency, and access conditions; contemporaneous judgment or intervention; and linked execution evidence. Wider accountability claims also need appeal, correction, repair, and reform records. Missing or deleted records and the applicable retention rule should be identified. Not every artifact is mandatory in every case, and collection must be proportionate to purpose and privacy risk.

Selected governance instruments suggest where such records might be relevant (Table 7). This is an interpretive crosswalk, not a compliance test. The historical military cases are not contemporary AI Act compliance assessments. The policy sources were reviewed at separate 2026 checkpoints; this editorial draft has not updated their

legal status.

The EU references are Regulation 2024/1689 and the separately recorded 2026 amendment and consolidated article pages. NIST references GOVERN 2.1, MAP 3.4, and MAP 3.5 of the 2023 framework, not an unexamined revision. Supplement Section S5 retains the policy dates, source-access qualifications, and selected New York City, California, Colorado, and Illinois comparisons. These address different actors and rights. Notice, bias audits, consumer appeals, and an overseer's event-level capabilities cannot be treated as interchangeable evidence of human control.

Post-cutoff context. Published on 2 November 2026, NIST IR 8621 records workshop participants' concerns about incomplete investigative evidence, provenance, fragmented authority across AI value chains, and preserving human decision points and downstream traces (Raunak et al., 2026, sections 4.2, 7, and 8.3). These discussions motivate further study of inspectable control evidence; they do not validate the present procedure or establish transfer from these historical cases to contemporary AI incident response. The report is a workshop synthesis, not a conformance standard. Its possible future NIST activities, including traceability and observability resources, are prospective rather than already-issued requirements (section 10). This contextual addition is outside the frozen literature search and historical packets and changes no assessment state or result. No NIST conformity or compliance claim is made.

7.2 A timing measure to test prospectively

When complete timestamps and durations are available, a proposed sequential intervention margin is

$$M_t = (t_{\text{commit}} - t_{\text{access}}) - (t_{\text{interpret}} + t_{\text{decide}} + t_{\text{transmit}} + t_{\text{propagate}}).$$

The first two terms are timestamps; the remaining terms are elapsed durations in the same unit. A nonnegative margin is necessary under this sequential measure but does not establish other feasibility conditions. Missing inputs leave the result Indeterminate. Parallel activity, interruption, uncertain endpoints, and unobserved durations require explicit treatment. The historical packets do not permit this calculation, and the measure is unvalidated.

7.3 Separate validation questions

Four studies would answer different questions. First, controlled conditions with known differences could test the procedure and artifact pipeline, including harmful but faithfully executed actions, ignored corrective instructions, missing evidence, and actual necessary-capacity failures. An authored reference key tests the specified behavior, not real-world adequacy.

Second, independent assessors should code the same frozen packets while blinded to earlier labels. The sampling plan, treatment of alternatives, agreement statistic, and adjudication rule should be fixed beforehand. Condition-level agreement is necessary alongside case totals: several assessors can all return Unresolved for different reasons. Third, construct and criterion validity require varied systems and tasks and an external criterion not derived from the six labels. Fourth, institutional benefit requires a defined outcome, such as correction or harm prevention; a causal claim needs a design capable of identifying a causal effect.

Participant recruitment or protected operational data require the applicable research-ethics determination, access and publication authority, consent or waiver basis, and data minimization, security, retention, and deletion arrangements before collection. A solo synthetic rehearsal can test executability after an appropriate recorded determination; it cannot establish independent reliability or institutional effects.

Later Jev work is a **separate synthetic companion**, not a validation of these historical rows. It used authored documentary-style stimuli with different question combinations, an authored key, exposure limitations, and no completed independent-human or unassisted-author comparison. The original first-run counts, invalid-response handling, and later assistance-pilot limits remain in Supplement Section S6. No new Jev run or experiment was conducted for this draft.

Table 7. Bounded relationship to selected governance instruments

Instrument	Reviewed function	Research use and limit
EU AI Act Article 14	Understanding limits, automation-bias awareness, interpretation, override, intervention, and stopping for high-risk AI.	Examine selected capabilities and the evidence of their event-level use; no legal-sufficiency finding.
EU AI Act Article 57	Supervised development, testing, and validation.	A possible prospective setting, subject to admission, permissions, and safeguards; none is established here.
NIST AI RMF 1.0	Documented roles, proficiency processes, and oversight arrangements.	Connect voluntary organization-level guidance to event records.
ISO/IEC 42001 public overview	AI management-system governance, risk, traceability, and improvement.	Context only; the licensed standard was not reviewed clause by clause, and no conformity or certification claim follows.

8 Limitations

Selection was purposeful and stopped after five candidates. The strata anticipated the headline contrasts, two cases concern one system family and period, and all three differ from present learned systems. The cases show procedure execution under selected evidence conditions, not prevalence, accuracy, or transfer.

The packets are incomplete retrospective public records. Original logs, displays, classified materials, and some full inquiries were unavailable. Source agreement can reflect common dependence. Remote content and preserved bytes have different evidentiary status. The method cannot recover records it does not possess.

One assessor designed and applied the method with substantive AI assistance. No independent human coding, inter-rater reliability, construct-validity estimate, safety effect, or outcome comparison is established. Oko's earlier six-label correction used a changed evidentiary interpretation, so it is not a same-rule stability test. Author approval and repeated automated critique are not independent validation.

The post hoc reassessment retains six unresolved P/I choices. It does not establish that adverse conditions were absent, and the original treatment of retrospective evidence for P remains incompletely specified. The 64-assignment invariance depends on the retained state sets. It cannot prove that those sets are warranted, and any justified U still requires Fail.

The literature review retains 980 missing retrieval outcomes and unresolved authenticated or disciplinary coverage, including Scopus or Web of Science, IEEE Xplore, ACM Digital Library, PhilPapers, and HeinOnline or an equivalent legal index. Five recovered forward sources have only bounded proposition permission; two remain background-only, six are quarantined, and RS-DQ-004 has none. Substantive overlap with existing methods and incomplete coverage preclude an exhaustive originality claim. The governance crosswalk establishes no legal sufficiency, certification, or regulatory endorsement.

9 Conclusion

Human control is not established by naming a reviewer. Under this procedure, an event-level claim is assessed by linking assigned authority to usable information, comprehension capacity, a feasible opportunity, active judgment or intervention, and propagation into execution. This procedure makes those links and their evidentiary limits inspectable.

The preserved baseline produced two Fail results and one Unresolved result. A limited correction gives one Fail and two Unresolved. The additional ZG710 reassessment gives three Unresolved results across 64 retained assignments, conditional on the approved judgments and other limited-correction cells. These are changes in warranted

documentary conclusions, not changes in historical control. A justified necessary-condition failure always retains priority.

The study's contribution is a method demonstration and a record of correction. Independent assessors, varied case families, contemporary traces, and prospective studies are needed before claims of reliability, validity, or institutional benefit are warranted.

Version and materials

The historical paper baseline is repository v0.17.0 at commit `781f7806a626cd4e4fffbe4d453d9d5365b835f9`. The public repository preserves protocols, packets, assessments, searches, figures, audits, and manifests. The evidence archive v0.6.0 is 10.5281/zenodo.21865007; the v0.14.0 preprint is 10.5281/zenodo.21926005; the all-versions repository DOI is 10.5281/zenodo.21841127. Citation metadata separately identifies 10.5281/zenodo.22970968 for the v0.17.0 paper. The author-approved 10 October revision and additional-reassessment supplement were published on 16 October at 10.5281/zenodo.23423847. The preceding 1 November editorial revision is archived at 10.5281/zenodo.23852566.

Later v0.17.1 and v0.17.2 software and synthetic additions did not silently revise the historical packets or paper. The 1 November editorial revision changed wording and disclosures, not states or results. The 8 November editorial revision reorganizes the paper and supplement while preserving the earlier findings and uncertainties. The author approved this revision for arXiv submission.

Ethics and author information

The historical study uses public records and involved no participant recruitment, intervention, or direct interaction with living individuals. Eligible public-domain and openly licensed materials are preserved; proprietary materials are cited through metadata and limited paraphrase. The protocol excludes confidential, restricted, and security-sensitive records. No institutional human-subjects-review determination has been obtained, and no exemption is claimed. This account does not establish an institutional or venue determination.

Node & Norm is the author's independent research initiative. Mark Julius Banasihan is an ALB candidate in Extension Studies at Harvard University. The research was conducted independently through Node & Norm, separate from his degree requirements and without Harvard University sponsorship, supervision, endorsement, or representation.

Author contributions. Mark Julius Banasihan: conceptualization, methodology, investigation, data curation, software, validation, visualization, drafting, review and editing, project administration, and accountable authorship. "Validation" denotes the specified artifact and consistency checks, not independently established instrument validity. **Conflicts of interest.** The author declares no conflicts of interest.

AI assistance and review limits

Generative AI assisted substantively with research preparation, interpretation, and manuscript development, beyond proofreading. Documented activities include literature discovery and metadata organization; source inspection, screening and proposition proposals; research and revision specifications; assessment-rule formulas, tables, figures, deterministic builders, claim maps, validation code; policy-source review; and drafting, alternative interpretations, and editorial revision. The v0.14 and v0.16 review records specify activities and limits. This editorial revision was also prepared with AI assistance under the author's review and authorization.

The inspected records identify Codex (OpenAI) for source inspection, manuscript and code preparation, and checks. They do not establish the exact underlying model version for every task or a complete inventory of earlier sessions; none is inferred. The separate synthetic experiment used TypeSafe AI's Jev, with requested and resolved `jev-1.13.0` and SDK 0.7.1 in its first live run. That identity does not identify the manuscript-development assistant.

Released records identify Mark Julius Banasihan as the accountable decision owner for the original 89-record AI-assisted queue and attribute

assessment and evidence approvals to him. Later work was assisted under his authorization. The v0.14 record states that he did not personally reproduce every assisted retrieval step. Authorization, recorded review, specific-claim verification, and release approval have different scopes. The approved additional ZG710 interpretation leaves P/I alternatives unresolved. The author remains responsible for adopted judgments, errors, and submitted text; AI tools are not authors.

Source inspection, static checks, executed tests, model inference, and human review are different activities. Hashes identify bytes, and passing checks establishes only specified behavior. Automated critiques, Jev outputs, and assisted author discussion are not independent human coding or peer review. No independent human validation of the method is established. Public records preserve selected activities, locators, decisions, and versions, not a complete transcript of manuscript development; the separate Jev archive preserves its requests and responses. Private conversation identifiers and unrelated private material are excluded. Any additional excerpts require factual, relevance, and privacy review. Venue-specific disclosure requirements remain to be reconciled before journal submission.

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